

# Novel marine antifouling coatings inspired by corals

L. Tian<sup>a, b</sup>, Y. Yin<sup>a</sup>, H. Jin<sup>a, \*</sup>, W. Bing<sup>a, c</sup>, E. Jin<sup>a</sup>, J. Zhao<sup>a</sup>, L. Ren<sup>a</sup>

<sup>a</sup> Key Laboratory of Bionic Engineering, Ministry of Education, Jilin University, Changchun, 130022, China

<sup>b</sup> Weihai Institute for Bionics, Jilin University, Weihai, 264207, China

<sup>c</sup> School of Chemistry and Life Science, Changchun University of Technology, Changchun, 130012, China

## ARTICLE INFO

### Article history:

Received 15 February 2020

Received in revised form

19 April 2020

Accepted 4 May 2020

Available online 24 May 2020

### Keywords:

Bioinspired antifouling coatings

Corals

Marine biofouling

Chemical strategy

Physical strategy

## ABSTRACT

Biofouling is a major problem facing the marine industry. Since toxic antifouling coatings were banned globally due to their negative impacts on the marine environment, the development of environmental-friendly and efficient antifouling coatings has been identified as a pressing need. As an alternative, the antifouling coatings inspired by corals have attracted a great deal of attention over these years. Within the marine environment, corals have evolved an excellent antifouling capability. There are five major antifouling strategies applied by corals, including natural antifoulants, foul release effect, sloughing effect, soft tentacles, and fluorescence effect. In this paper, a brief review is conducted to introduce the antifouling coatings inspired by the five strategies. Moreover, a discussion is conducted about the existing problems with the five strategies and the direction of their further development is indicated.

© 2020 Elsevier Ltd. All rights reserved.

## 1. Introduction

Biofouling refers to the attachment of undesirable molecules and fouling organisms to the submerged surfaces [1]. It tends to have a significant negative impact on marine vessels since the attached fouling organisms could play a major role in exacerbating surface corrosion and increasing drag resistance. As a consequence, it incurs a greater amount of maintenance cost and increases fuel consumption [2]. In addition, biofouling can lead to bio-invasion if fouling organisms travel with marine vessels [3]. For a variety of different underwater marine facilities, such as net cages for aquaculture, cross-sea bridge, and oil production platform, biofouling can cause damage to these facilities, which may result in substantial economic losses [4]. In general, the conventional antifouling coatings contain toxic substances, for example, arsenic, sulfur, mercury, and TBT [5]. These toxic antifouling coatings tend to have unintended detrimental effects on marine organisms and are capable to deposit and accumulate inside the body of marine creatures. Eventually, it could gain entry into the human body through food chain, thus causing harm to human health [6]. Back in the 1980s, people were made aware of these problems, and those toxic antifouling coatings since were gradually prohibited from use

worldwide [7]. In 1999, the International Maritime Organization (IMO) passed an assembly resolution to impose ban on the use of TBT by 1 January 2003 [8]. Thus, it is absolutely necessary to develop a sort of environmental-friendly and efficient antifouling coatings.

At present, there are three major environmental-friendly antifouling strategies, which are foul release coatings (FRCs) [9,10], protein resistance coatings (PRCs) [11,12], and bioinspired coatings [13,14]. FRCs have attracted a great deal of attention in recent years, as they are capable to be commonly fabricated with such low surface energy materials as silicone, which is environmental-friendly, efficient, and cheaply affordable. However, FRCs require a high speed to remove some varieties of fouling organisms [15], and it is difficult for them to bond to the hull surface due to their low surface energy. As for PRCs, they cannot maintain the antifouling capability in case of long-term use [4]. These drawbacks restrict their practical applications in the marine industry. As an alternative to the traditional antifouling coatings, bioinspired strategy has drawn a great deal of attention over these years, and plenty of efforts have been made to explore their potential to reduce biofouling. Marine organisms have developed a variety of different antifouling strategies to resist biofouling. For example, shark get rid of biofouling through their skin denticles [16], mussels are capable to resist biofouling through their textured shells [17], and mangrove trees *Sonneratia apetala* prevent biofouling through their micro-textured leaves [18]. These features contribute to a well-known antifouling

\* Corresponding author.

E-mail address: [ibaidu@foxmail.com](mailto:ibaidu@foxmail.com) (H. Jin).

strategy, that is, micro-textured surfaces [19]. Over the past few decades, micro-textured surfaces prompted plenty of studies. However, it has been suggested in some studies that fouling organisms can gradually fill up the valleys, thus masking the microstructures, as a consequence of which coatings could lose their antifouling capability [20]. Besides, the microstructures are susceptible to damage even by gentle pressing or wiping of finger [21]. As revealed by a recent study, microstructures have no effect or inconclusive effect on fouling in 46% of the cases involved [19]. These potential drawbacks present a tough challenge for their engineering applications. Therefore, the development of new bio-inspired antifouling strategies is considered as necessary.

Recently, researchers have conducted investigation into antifouling strategies as developed by corals as they usually have an active and clean surface [22–25]. As corals are classed as sessile organisms, and they are immobile even under the context of external stimuli, it is a necessity for them to develop and apply special antifouling strategies. Antifouling coatings inspired by these strategies can provide protection for moored vessels and static facilities within an underwater environment. It has been demonstrated in previous studies that corals have developed five major antifouling strategies, including antifoulants, foul release surface, sloughing slime, soft tentacles, and fluorescent effect (Fig. 1).

- 1. Natural antifoulants:** natural antifoulants are extracted from several marine organisms (e.g., macroalgae, sponges, and soft corals). They can effectively reduce the proliferation of fouling organisms [26]. Soft corals refer to one of the most prolific sources of bioactive marine natural products [22]. Thus, antifoulants from soft corals have aroused huge attention these years compared with that from other marine species.
- 2. Foul release effect:** some varieties of coral skins are demonstrated with low surface energies, thus which allow them to exhibit foul release property [27,28]. Some organisms (e.g., sharks, dolphins, and lotus) have been found to resist biofouling via foul release effect [19,29]. The mentioned organisms combat biofouling via their micro/nano structures on their surfaces. Unlike these species, corals exhibiting low surface energy

prevent biofouling in a simple manner, i.e., the lowest adhesion strength; in other words, the surface energy of coral skin is at the bottom of the Baier curve [27].

- 3. Sloughing effect:** the corals are capable of removing fouling organisms via a slime sloughing process [30,31], similar to taking off old clothes and putting on new ones. This process is a nearly perfect approach to resist approximate 100% biofouling compared with other strategies developed by other species.
- 4. Soft tentacles:** the coral tentacles swing in the flow to resist biofouling. Our recent study found that corals have adopted additional physical antifouling strategies. Such soft coral as *Sarcophyton trocheliophorum* develops soft tentacles on their surfaces to swing in the water flow; as a result, an unstable surface is formed around their body [32]. Accordingly, the fouling organisms are prevented from making contact with their surface, while the attached fouling organisms can easily be removed. This antifouling strategy refers to a physical process, indicating its green, durable, and efficient characteristic.
- 5. Fluorescent effect:** fluorescence is capable of resisting attachment of diatoms. The fluorescent corals are capable of emitting weak light to provide energy for photosynthesis of zooxanthellae. Moreover, the weak light can effectively inhibit the settlement of diatoms, so the antifouling capability is achieved [33]. Thus far, no other marine species with fluorescence have been reported to exhibit antifouling capability. For this reason, such strategy may act as a unique antifouling strategy developed by fluorescent corals.

These strategies are deemed novel and interesting. The antifouling coatings inspired by these strategies are characterized by environmental friendliness, efficiency, and low cost. In this paper, the focus is placed on the progress made in the five strategies applied by corals and discussion around the existing problems. Moreover, the prospect of future development for the antifouling coatings inspired by corals is indicated.

## 2. Results and discussion

### 2.1. Natural antifoulants from corals

Corals inhabit tropical waters around the world. They have developed the antifouling capability to get rid of biofouling. Up to now, more than 5,800 varieties of compound have been isolated from corals [34]. Many of these compounds have been investigated to reveal their potential to combat biofouling. These compounds feature environmental friendliness and efficiency. On average, there are more than 100 published works of literature concerning the antifoulants isolated from corals annually, with the details presented in Fig. 2a. From literature, it can be found out that there have been over 52 coral species and 188 compounds reported over the recent decade. The pie chart (Fig. 2b) indicates the percentage of different classes of compounds and crude extracts derived from corals in the most recent decade. These compounds include sesquiterpenes, steroids, alkaloids, and diterpenes [35–38]. For example, cembrane diterpenes (–)14-deoxycrassin ( $EC_{50}$  for the bryozoan *Bugula neritina* 3.90  $\mu\text{g/ml}$ ) [22] and sarcoglaucin B ( $EC_{50}$  for the barnacle *Balanus Amphitrite* 7.52  $\mu\text{g/m}$ ) [24] exhibit remarkable capability to inhibit the proliferation of fouling organisms. According to the U.S. Navy Program, a compound with  $EC_{50} < 25 \mu\text{g/ml}$  can be considered as a promising antifouling antifoulant [39]. Additionally, the antifouling coatings as derived from combining these compounds and inert paint also demonstrate excellent antifouling capability [23]. After 185 days of immersion in seawater, the hybrid coatings containing the extracts of two corals *Sinularia compressa* and *Heteroxenia fuscescens* exhibited

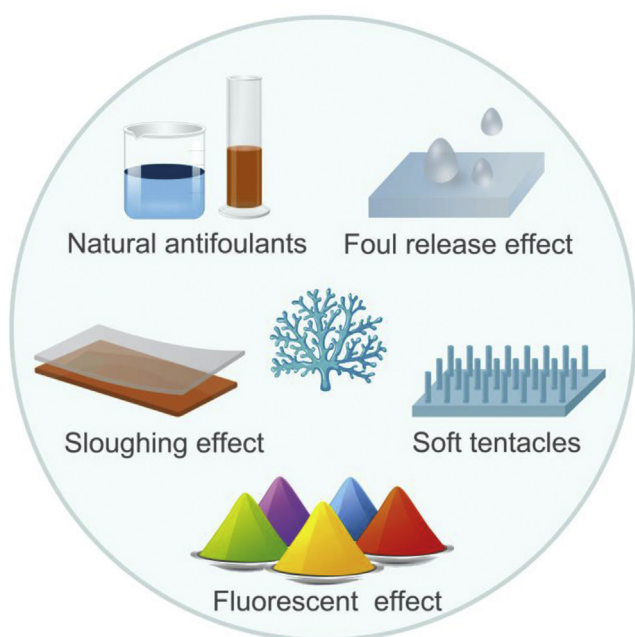
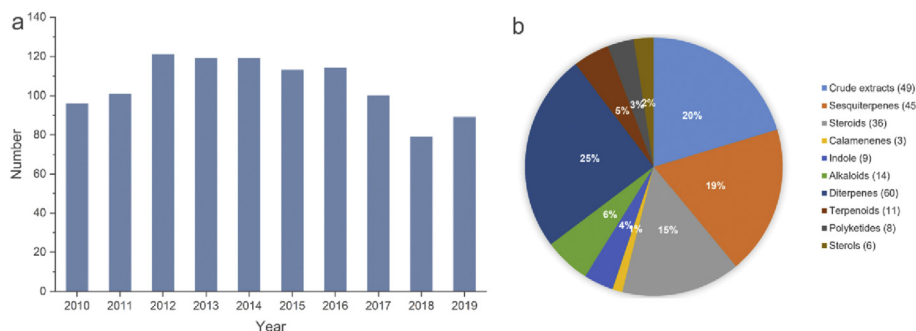


Fig. 1. Schematic illustration of five antifouling strategies inspired by corals.



**Fig. 2.** (a) The number of annually published research articles in relation to natural antifoulants from corals, from 2010 to 2019. The data was collected using Google Scholar tool with “antifoulants + coral” as the keywords. (b) Different classes of the compounds and crude extracts derived from corals in the most recent decade.

impressive antifouling capability against tubeworms and barnacles. These compounds have some already-known mechanisms, including toxins, growth inhibition, interference of neural pathways, antiadhesion, and repellants [40–43]. However, the antifouling mechanisms of most compounds remain unclear. It is speculated that some of them have multiple antifouling mechanisms [44].

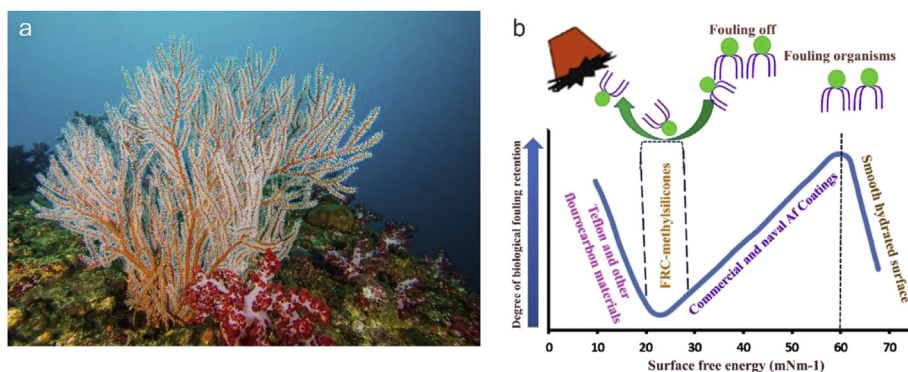
It has been discovered in research that not all of the compounds could show a great antifouling capability to resist biofouling, and a sort of compound alone is incapable of performing bio-activity to resist all varieties of the fouling organisms. Wang et al. [45] extracted four kinds of terpenoids from the soft coral *Simularia multiflora*. In the test against the barnacle *Balanus albicostatus*, only such compounds as diterpenoid multifloralin and sclerophytin E exhibited potential of antifouling activity. Zhang et al. [24] conducted study of 12 cembrane diterpenes from the soft coral *Sarcophyton glaucum*, only five of which showed strong antifouling activity against the barnacle *Balanus Amphitrite*. Unfortunately, none of these compounds showed antibacterial activity in the test against Gram-negative and Gram-positive bacteria. Similar results were also reported by Wang et al. [46], and Dobretsov et al. [25], which suggested that the compounds as derived from corals have no capability of broad-spectrum antifouling activity, thus presenting a challenge for their practical applications. Therefore, the use of multiple compounds in antifouling coatings or search for some new broad-spectrum compounds may provide a solution to this.

## 2.2. Antifouling coatings inspired by foul release effect

Gorgonian corals (Fig. 3a) features a clean surface, which indicates that they have antifouling capability. They have been found

to have surfaces with a low level of surface energy ( $\gamma = 23\text{--}27\text{ mN/m}$ ) [27] within the range of minimal adhesiveness of the prominent Baier curve (Fig. 3b), which evidences the association between surface energy and the relative strength of adhesion. The relative strength of adhesion between a fouling organism and a surface with 20–25 mN/m surface energy is at the minimal level, thus facilitating the fouling organism to be removed by the movement of vessels [47]. This physical antifouling mechanism inspires a novel type of antifouling coatings, known as FRCs. Over the past decades, the materials with low surface energies, such as fluoropolymers and silicone have been extensively explored to reveal their antifouling capability [1,48,49]. These materials are inert, which is contributory to their excellent stability within marine environment. Moreover, they rely on a physical process to achieve the antifouling effect, which features environmental friendliness. These desirable characteristics as mentioned above thus attract increasing attention for study over the past decades [4]. However, as referred to the introduction section, it is difficult for these surfaces with low surface energies to bond to vessels. Moreover, the mechanical strength of these materials is fairly low, which could cause the coatings to develop splits and cracks when exposed to the harsh environmental conditions. Therefore, these drawbacks severely restrict their practical applications in the marine environment.

Recently, a variety of nanofillers, such as  $\text{SiO}_2$  [50],  $\text{TiO}_2$  [10],  $\text{MnO}_2$  [51],  $\text{SiC}$  [51], and  $\text{ZnO}$  [52], were combined with silicone to improve its mechanical strength and antifouling capability. These silicone-based coatings have non-leaching property and resist biofouling through a physical process, which ensures environmental friendliness. In addition, only a small fraction of nanofillers contained in silicone could spring up a new coating with strong antifouling activity. In our recent research work [53], silicone



**Fig. 3.** (a) A colony of gorgonian coral (with permissions of use granted from [tuchong.com](http://tuchong.com)). (b) The association between surface energy and the relative strength of adhesion (adapted from Ref. [1] with permission from Elsevier © 2017).

rubber was chosen as substrate, and graphene nanosheets were introduced as fillers, which led to the finding that the coating containing 0.36 wt% graphene nanosheets exhibited excellent antifouling capability against bacteria *Paracoccus pantotrophus*. Then, a similar study was conducted by taking graphene oxide as fillers and silicone rubber as substrate [14]. The 0.36 wt% coating demonstrated remarkable antifouling capability against diatoms. Moreover, other sorts of carbon-based nanoparticles, such as carbon nano tubes (CNT), were also commonly used as fillers to enhance the mechanical strength [54,55]. In general, FRCs are faced with a promising prospect if the bond strength between FRCs and vessels is improved to the required levels.

### 2.3. Antifouling coatings inspired by sloughing effect

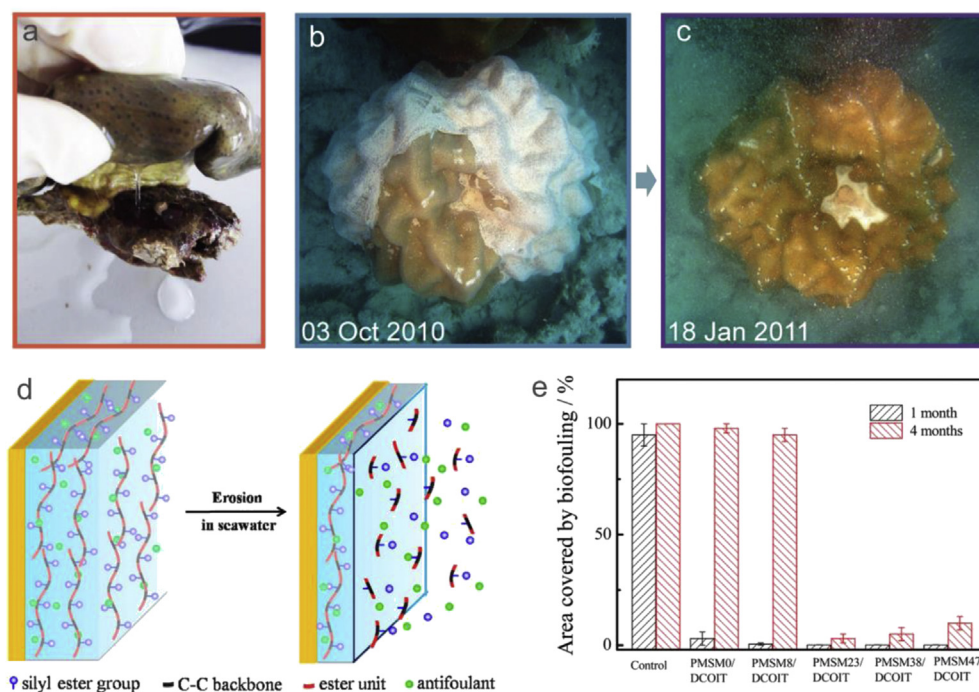
The coral mucus (Fig. 4a) is capable to protect corals from biofouling by taking different approaches, including a physical barrier, antimicrobial compounds, ingestion by the coral host, and slime sloughing effect [31,56]. Among these approaches, slime sloughing effect reduces biofouling through a mechanical process, which ensures efficiency and environmental friendliness. Fig. 4b shows a coral colony (*Porites spp.*) that is sloughing its mucous sheet, and it develop a fresh new surface in three months (Fig. 4c) [57]. The similar antifouling strategy was also observed in the alga, which is effective in preventing biofouling by means of cuticle peeling [58–60]. The antifouling mechanism of sacrificial coatings [61], self-polishing coatings [62], and degradable copolymers [12] shows similarity to the sloughing effect. For example, Wang et al. [63] fabricated a solid organogel coating with a regenerable sacrificial alkane surface layer. It is easy for the foreign fouling materials to be removed through the regenerable process, which shows a massive potential to combat biofouling. However, the authors failed to validate its antifouling capability. Dai et al. [12] reported a self-generating and self-renewing zwitterionic polymer surface, the

coatings of which could be effectively in resisting marine bacteria (*Pseudomonas sp.*) and diatoms (*Navicula incerta*).

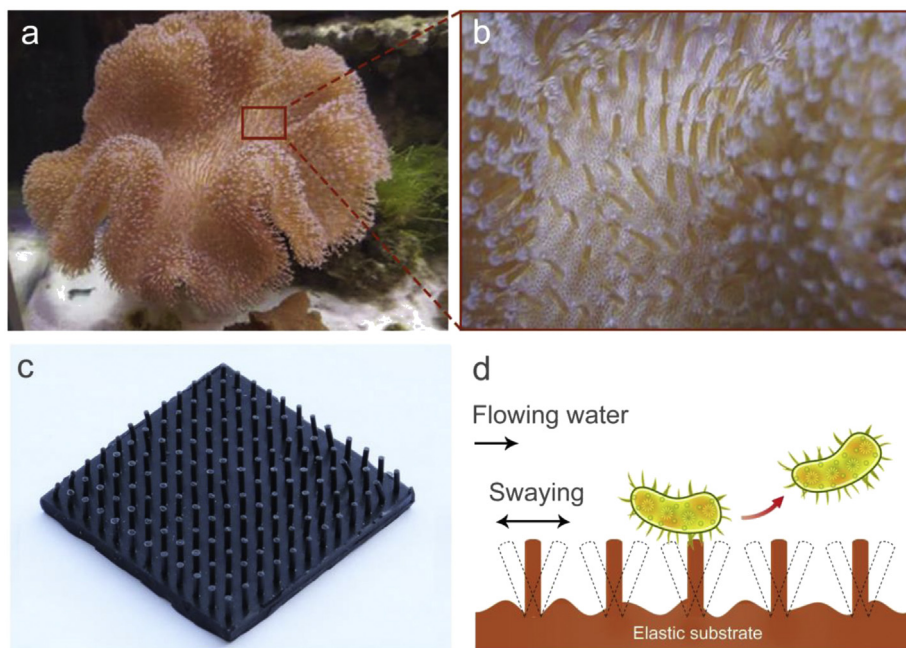
Apart from the above-mentioned coatings, polyurethane and acrylate copolymer have also been widely applied in the production of self-renewing and degradable antifouling coatings [64–67]. Fig. 4d illustrates the antifouling mechanism of a degradable acrylate antifouling coating [68], which is similar to the sloughing process. The antifouling capability is made controllable by the molar ratio of ester units in the backbone to the silyl ester side groups. Fig. 4e indicates the results of the samples with different molar ratios in marine field tests. It can be seen from the figure that the coatings with > 20% MDO content exhibited excellent antifouling capability after a four months of marine field test. Therefore, these degradable antifouling coatings are considered as having a promising future of practical applications on marine vessels. However, there are some issues in relation to this antifouling strategy. From the mechanism of self-renewing and degradable antifouling coatings, it can be found out that it is essential to exercise control over degradable speed. More specifically, if the degradable speed is high, the coatings would show remarkable antifouling capability but the lifespan could be reduced. Conversely, if the degradable speed is low, it is unlikely for the coatings to exhibit a good antifouling activity. Therefore, it is a challenge to determine an appropriate degradation speed for such coatings in practice.

### 2.4. Antifouling coatings inspired by coral tentacles

The soft coral *Sarcophyton trocheliophorum* contains plenty of soft tentacles on its body (Fig. 5a and b), which make it capable to swing in the flow of water. In our recent studies, it has been discovered that the swing effect of soft tentacles is among the antifouling strategies applied by the soft coral [32]. The tentacles constitute an unstable surface under flow, which prevents the fouling organisms from adhering to such a dynamic surface. Based on this strategy, the silicone with a low Young's modulus was taken



**Fig. 4.** (a) Mucus derived from a soft coral (*Sarcophyton trocheliophorum*). (b, c) A mucous sheet sloughing process developed for the coral *Porites spp.* (adapted and modified from Ref. [57]). (d) Antifouling mechanism of the acrylate coating. (e) Results of the marine field test (adapted and modified from Ref. [68] with permission from American Chemical Society © 2015).



**Fig. 5.** (a) The soft coral *Sarcophyton trocheliophorum*. (b) Soft tentacles of the soft coral (adapted and modified from Ref. [32] with permission from John Wiley and Sons © 2019). (c) Photo of the antifouling coating. (d) Antifouling mechanisms of the tentacles.

as the substrate to fabricate antifouling coatings, and graphene nanosheets were added into the silicone for improvement of its mechanical strength. The antifouling coatings were fabricated by using negative molds, with the appearance of the antifouling coating illustrated in Fig. 5c. In the attachment test, the 0.36 wt% coatings demonstrated the most desirable antifouling capability against Gram-negative (*Paracoccus pantotrophus*) and Gram-positive (*Bacillus subtilis*) bacteria. By performing a fluid-solid coupling simulation, we made a discovery that the tentacles are capable to swing with a 10 Hz frequency under the context of turbulent flow. These coatings could inhibit fouling organisms through a physical process (Fig. 5d), which makes them different from the mechanism of biocides. Therefore, they show such characteristics as environmental friendliness and durability. In theory, any sorts of soft materials have the potential to be applied in the fabrication of such coatings with tentacles. Moreover, these materials and fabrication procedures are easy to achieve and cheaply affordable, which presents these coatings with a promising future. Nevertheless, the evaluation of these coatings for antifouling capability was restricted to lab conditions. As the marine environment is highly complicated, it remains necessary for the marine field test to be conducted in the future. In addition, this work is a primary study. Therefore, further researches are required. For example, the relationships between tentacle properties (density, length, radius, and elastic modulus) and antifouling capability remain unclear.

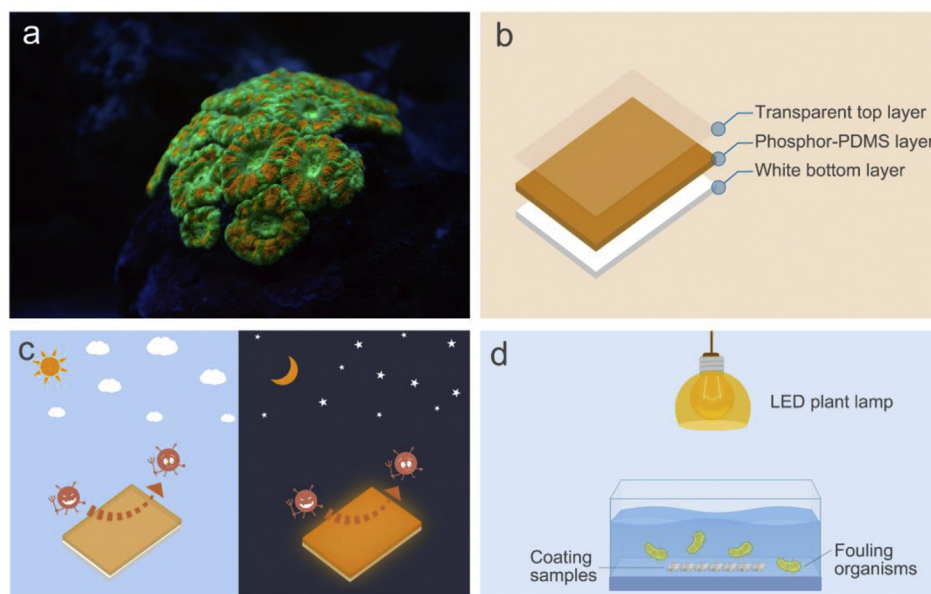
### 2.5. Antifouling coatings inspired by fluorescent corals

Not only do fluorescent corals contain fluorescent proteins to protect themselves [69], they are also capable of emitting weak light that enables the zooxanthellae to photosynthesize with the aim of food in exchange [70]. Fluorescent corals display a variety of different colors, including red, green, yellow, blue, and others (Fig. 6a). In prior literature, it has been reported that the weak light could inhibit the settlement of diatoms [71]. In consideration of this, an idea is suggested that the fluorescence emitted by corals

can be applied as one of the antifouling strategies. In our recent work [33], a sandwich-structure coating inspired by fluorescent corals was reported. The coating consists of three layers (Fig. 6b), which are the transparent top layer (PDMS), the middle layer (phosphor layer), and the bottom layer (silicone rubber). The middle layer can emit weak light in the dark after going through a lighting procedure, while the other two layers provide protection for the middle layer. These coatings were expected to provide antifouling capability throughout the day (Fig. 6c), as evidenced by the attachment test (Fig. 6d). In the adhesion assay, two algae (*Chlorella*, *N. closterium* f. *minutissima*) were selected as the fouling organisms. The coatings showed strong antifouling activity to combat these fouling organisms, which suggests their antifouling capability to prevent algal adhesion. Furthermore, the sandwich-structure coatings with different color phosphors were prepared and their antifouling activity was tested, which led to the finding that the inhibitory effect against algae was affected by the wavelength of the light emitted from the phosphor layer. More specifically, blue (427 nm), yellow (663 nm), and red (731 nm) phosphor layers showed strong anti-algae activity, while the green (512 nm) phosphor layer showed only insignificant effect on countering algae. As algae are photosynthetic organisms, their photosynthesis are affected by the wavelength of the light [71,72]. The antifouling mechanism of the sandwich-structure conforms to a physical process, which features environmental friendliness. In addition, the top layer is modifiable to promote the antifouling activity, as a result of which these coatings are believed to have a promising prospect of practical applications.

### 3. Summary and outlook

In general, the five antifouling strategies are characterized by environmental friendliness and efficiency. They demonstrate a massive potential of practical application in marine vessels. Besides, the mentioned strategies exhibit additional advantages. The benefits of different strategies are summarized and analyzed in the following:



**Fig. 6.** (a) Photo of the fluorescent coral. (b) Schematic diagram of the sandwich-structured coating. (c) The coatings perform antifouling activity for 24 h throughout the day. (d) Schematic diagram of the attachment testing (adapted and modified from Ref. [33] with permission from John Wiley and Sons © 2020).

1. **Natural antifoulants:** some natural antifoulants from corals exhibit very low  $EC_{50}$ , indicating that they can effectively combat biofouling. Also, natural antifoulants can be incorporated into materials to fabricate antifouling paints and coatings. Compared with other bioinspired strategies, such strategy can be combined with more materials.
2. **Foul release effect:** this strategy is dependent of the intrinsic nature of materials; in other words, the surface energy of materials to reduce biofouling is of critical significance. Hence, any materials exhibiting appropriate surface energy can be exploited to fabricate such antifouling coatings. For instance, the silicone refers to a very cheap material, so its production is low-cost. Moreover, this strategy releases no substances into seawater, so it is considered a greener strategy than natural antifoulants and sloughing coatings.
3. **Sloughing effect:** theoretically, similar to putting off dirty clothes, this strategy is capable of removing all the fouling organisms in the sloughing process, suggesting that such strategy exhibits high effectiveness. Besides, the antifouling capability and life span of this coating are adjustable.
4. **Soft tentacles:** the fouling organisms cannot easily adhere to the soft tentacles in the water flow. Thus, the soft tentacles can enhance the removal efficiency of a common material. For this reason, any suitable soft materials exhibit the potential to fabricate such coatings.
5. **Fluorescent effect:** the sandwich structures we reported exhibit a considerable number of advantages. It exhibits prominent extendibility compared with other strategies. For instance, the top layer can be modified with natural antifoulants, microstructures, foul release materials, and protein resistance materials.

Despite these advantages mentioned above, the five strategies are still subject to limitations. The five strategies can be classified into two different types, i.e., chemical and physical approaches.

1. **Natural antifoulants:** natural antifoulants are derived from taking chemical approaches, so that the challenge lies in the

lifespan of the antifoulants. Another drawback is that a large majority of the antifoulants are incapable of broad-spectrum antifouling activity, which could restrict their applications. Therefore, it is necessary to conduct search for broad-spectrum antifoulants in the future. Moreover, the corals are ecologically sensitive, and coral reefs are reducing [73]. Accordingly, the overdeveloping of natural antifoulants from corals may damage the coral ecosystem.

2. **Foul release effect:** as mentioned in the introduction section, the less-than-satisfactory bond strength adversely affects their practical applications. If this problem could be resolved, it is a real possibility to make a significant progress in this regard. Moreover, a high speed is required to remove some kinds of fouling organisms for a marine vessel. Thus, the application of this strategy will be restricted under low hydrodynamic conditions.
3. **Sloughing effect:** the sloughing effect involves chemical and physical antifouling mechanisms, which also causes some problems to be resolved, such as the control of suitable degradation speed and lifespan. The environmental risk of this strategy should be fully assessed.
4. **Soft tentacles:** these studies remain preliminary, suggesting that in-depth studies are required. Furthermore, the massive production may be restricted for the existing preparation methods.
5. **Fluorescent effect:** the fluorescence exhibits a specificity to resist algal adhesion. As it has no effect on bacteria, it should be combined with other strategies to achieve a broad-spectrum antifouling capability. Besides, the phosphor layer requires sunlight to store energy in the day and emits fluorescent light at night. Given this, the antifouling activity may be reduced in the cloudy and rainy days.

On the whole, the antifouling coatings should be environment-friendly, low-cost, and efficient. In accordance with the mentioned principle, some of the strategies developed by corals have not been appropriately applied. For instance, the massive isolation of natural antifoulants from corals is not practical, and the collection of corals

may impair the coral ecosystem. Accordingly, the synthesis of similar compounds may be a solution. The studies on soft tentacles and fluorescent effect remain preliminary, and never find their way outside the lab. Besides, the massive production is restricted in the current preparation methods, which may bring a poor economic performance. 3D printing refers to an advanced technology to yield complex products. It has numerous advantages (e.g., low-cost, faster production, and easy design). Thus, 3D printing may be a promising production method to fabricate such soft tentacles. The fluorescent effect fails to resist bacteria, and thus it should be combined with other strategies. Over the past few years, studies have been conducted extensively on antifouling coatings based on foul release effect and sloughing effect. They are low-cost, efficient, and convenient for large-scale production. In our opinion, they are the most promising strategies to be applied on marine vessels. However, there are challenges for their applications. The poor bonding strength of foul release coatings to vessels should be effectively improved. The environmental risk of coatings based on the sloughing effect should be fully assessed.

Given the desirable requirements altogether, it remains a challenge for a single strategy to realize a perfect antifouling capability. On the whole, no single strategy can exhibit prominent antifouling capability in all situations. Thus, the combination of various strategies is expected to lead to spring up a superior antifouling coating. The different strategies can exert a synergistic effect to combat biofouling. In the multi-strategy antifouling coatings, if one of the strategies lose efficacy, the other strategies can still be effective. Hence, the robustness and durability of the antifouling coatings can be improved. Currently, antifouling coatings based on multi strategies have been rarely reported, and thus more efforts are required in the future.

Overall, antifouling coatings inspired by corals are highly promising and face significant challenges. They are environment-friendly, low-cost, and efficient, which have aroused increasing attention these years. Though there exist some drawbacks, they are expected to be well remedied as fueled by the advancement of material science and advanced manufacturing technology. We believe that the antifouling coatings based on multi strategies inspired by corals have a promising future.

## Declaration of Competing Interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

## Acknowledgements

This work was funded by the National Natural Science Foundation of China General Program (Grant No. 51875240), the Fund Project in Equipment Pre-research Field (Grant No. 61400040403), the China Postdoctoral Science Foundation funded project (Grant No. 2018M630324), the Special Industrialization Demonstration project "New Material Special Project" of Jilin province and school co-construction plan (Grant No. SXGJSF2017-3), the Major Scientific and Technological Project of Changchun (Grant No.17SS023), and the Department of Science and Technology of Jilin Province (Grant No. 20190103114JH).

## References

- [1] M.S. Selim, M. Shenashen, S.A. El-Safty, S. Higazy, M.M. Selim, H. Isago, A. Elmarakbi, Recent progress in marine foul-release polymeric nanocomposite coatings, *Prog. Mater. Sci.* 87 (2017) 1–32.
- [2] E. Almeida, T.C. Diamantino, O. de Sousa, Marine paints: the particular case of antifouling paints, *Prog. Org. Coating* 59 (2007) 2–20.
- [3] A. Lacoursière-Roussel, D.G. Bock, M.E. Cristescu, F. Guichard, C.W. McKindsey, Effect of shipping traffic on biofouling invasion success at population and community levels, *Biol. Invasions* 18 (2016) 3681–3695.
- [4] Q. Xie, J. Pan, C. Ma, G. Zhang, Dynamic surface antifouling: mechanism and systems, *Soft Matter* 15 (2019) 1087–1107.
- [5] J.A. Lewis, Battling biofouling with, and without, biocides, *Chem. Aust.* (2018) 26–29.
- [6] B. Antizar-Ladislao, Environmental levels, toxicity and human exposure to tributyltin (TBT)-contaminated marine environment. A review, *Environ. Int.* 34 (2008) 292–308.
- [7] M. Adnan, E. Alshammari, M. Patel, S.A. Ashraf, S. Khan, S. Hadi, Significance and potential of marine microbial natural bioactive compounds against biofilms/biofouling: necessity for green chemistry, *PeerJ* 6 (2018) 5049.
- [8] R.M. Batista, I.B. Castro, G. Fillmann, Imposed and butyltin contamination still evident in Chile after TBT global ban, *Sci. Total Environ.* 566 (2016) 446–453.
- [9] M.S. Selim, H. Yang, F.Q. Wang, N.A. Fathallah, X. Li, Y. Li, Y. Huang, Superhydrophobic silicone/SiC nanowire composite as a fouling release coating material, *J. Coating Technol. Res.* 16 (2019) 1165–1180.
- [10] M.S. Selim, S.A. El-Safty, A.M. Azzam, M.A. Shenashen, M.A. El-Sockary, O.M. Abo Elenien, Superhydrophobic silicone/TiO<sub>2</sub>-SiO<sub>2</sub> nanorod-like composites for marine fouling release coatings, *Chemistry* 4 (2019) 3395–3407.
- [11] C.D. Beyer, M.L. Reback, S. Gopal, K.A. Nolte, J.A. Finlay, A.S. Clare, L. Schäfer, N. Metzler-Nolte, A. Rosenhahn,  $\alpha$ -Aminoisobutyric acid stabilized peptide SAMs with low nonspecific protein adsorption and resistance against marine biofouling, *ACS Sustain. Chem. Eng.* 8 (2020) 2665–2671.
- [12] G. Dai, Q. Xie, X. Ai, C. Ma, G. Zhang, Self-generating and self-renewing zwitterionic polymer surfaces for marine anti-biofouling, *ACS Appl. Mater. Interfaces* 11 (2019) 41750–41757.
- [13] Z. Li, Z. Guo, Bioinspired surfaces with wettability for antifouling application, *Nanoscale* 11 (2019) 22636–22663.
- [14] H. Jin, W. Bing, L. Tian, P. Wang, J. Zhao, Combined effects of color and elastic modulus on antifouling performance: a study of graphene oxide/silicone rubber composite membranes, *Materials* 12 (2019) 2608.
- [15] K.Z. Hunsucker, H. Gardner, K. Lieberman, G. Swain, Using hydrodynamic testing to assess the performance of fouling control coatings, *Ocean. Eng.* 194 (2019) 106677.
- [16] L. Wen, J.C. Weaver, G.V. Lauder, Biomimetic shark skin: design, fabrication and hydrodynamic function, *J. Exp. Biol.* 217 (2014) 1656–1666.
- [17] X. Bai, G. Xie, H. Fan, Z. Peng, C. Yuan, X. Yan, Study on biomimetic preparation of shell surface microstructure for ship antifouling, *Wear* 306 (2013) 285–295.
- [18] J. Fu, H. Zhang, Z. Guo, D.-q. Feng, V. Thiyagarajan, H. Yao, Combat biofouling with microscopic ridge-like surface morphology: a bioinspired study, *J. R. Soc. Interface* 15 (2018) 20170823.
- [19] M. Carve, A. Scardino, J. Shimeta, Effects of surface texture and interrelated properties on marine biofouling: a systematic review, *Biofouling* 35 (2019) 597–617.
- [20] A. Bers, E. Díaz, B. Da Gama, F. Vieira-Silva, S. Dobretsov, N. Valdivia, M. Thiel, A. Scardino, C. McQuaid, H. Sudgen, Relevance of mytilid shell microtopographies for fouling defence—a global comparison, *Biofouling* 26 (2010) 367–377.
- [21] D. Zhang, L. Li, Y. Wu, B. Zhu, H. Song, One-step method for fabrication of bioinspired hierarchical superhydrophobic surface with robust stability, *Appl. Surf. Sci.* 473 (2019) 493–499.
- [22] J. Wang, P. Su, Q. Gu, W.D. Li, J.L. Guo, W. Qiao, D.Q. Feng, S.A. Tang, Antifouling activity against bryozoan and barnacle by cembrane diterpenes from the soft coral *Sinularia flexibilis*, *Int. Biodeterior. Biodegrad.* 120 (2017) 97–103.
- [23] Y.A.A. Soliman, A.M. Ibrahim, A.H. Moustafa, M.A.F. Hamed, Antifouling evaluation of extracts from Red Sea soft corals against primary biofilm and biofouling, *Asian Pac. J. Trop. Biomed.* 7 (2017) 991–997.
- [24] J. Zhang, X. Tang, X. Han, D. Feng, X. Luo, L. van Ofwegen, P. Li, G. Li, Sarcoglaucins A1, new antifouling cembrane-type diterpenes from the South China Sea soft coral *Sarcophyton glaucum*, *Org. Chem. Front.* 6 (2019), 2004–2013.
- [25] S. Dobretsov, A.S. Al-Wahaibi, D. Lai, J. Al-Sabahi, M. Claereboudt, P. Proksch, B. Soussi, Inhibition of bacterial fouling by soft coral natural products, *Int. Biodeterior. Biodegrad.* 98 (2015) 53–58.
- [26] H.U. Dahms, S. Dobretsov, Antifouling compounds from marine macroalgae, *Mar. Drugs* 15 (2017) 265.
- [27] N. Vrolijk, N. Targett, R. Baier, A. Meyer, Surface characterisation of two gorgonian coral species: implications for a natural antifouling defence, *Biofouling* 2 (1990) 39–54.
- [28] A. Clare, D. Rittschof, D. Gerhart, J. Maki, Molecular approaches to nontoxic antifouling, *Invertebr. Reprod. Dev.* 22 (1992) 67–76.
- [29] A.P. Malshe, S. Bapat, K.P. Rajurkar, H. Haitjema, Bio-inspired textures for functional applications, *CIRP Ann* 67 (2018) 627–650.
- [30] L.A. Gomez-Lemos, G. Diaz-Pulido, Crustose coralline algae and associated microbial biofilms deter seaweed settlement on coral reefs, *Coral Reefs* 36 (2017) 453–462.
- [31] K.B. Ritchie, Regulation of microbial populations by coral surface mucus and mucus-associated bacteria, *Mar. Ecol. Prog. Ser.* 322 (2006) 1–14.
- [32] W. Bing, L. Tian, Y. Wang, H. Jin, L. Ren, S. Dong, Bio-inspired non-bactericidal coating used for antibiofouling, *Adv. Mater. Technol.* US 4 (2019) 1800480.
- [33] H. Jin, W. Bing, E. Jin, L. Tian, Y. Jiang, Bioinspired PDMS–phosphor–silicone rubber sandwich-structure coatings for combating biofouling, *Adv. Mater. Interfaces* 7 (2020) 1901577.

- [34] G. Li, P. Li, X. Tang, Natural products from corals, in: *Symbiotic Microbiomes of Coral Reefs Sponges and Corals*, Springer, 2019, pp. 465–504.
- [35] S.-H. Qi, X. Ma, Antifouling compounds from marine invertebrates, *Mar. Drugs* 15 (2017) 263.
- [36] M.-E.F. Hegazy, T.A. Mohamed, A.I. Elshamy, M.A. Al-Hammady, S. Ohta, P.W. Paré, Casbane diterpenes from red sea coral *Sinularia polydactyla*, *Molecules* 21 (2016) 308.
- [37] C.-L. Shao, R.-F. Xu, C.-Y. Wang, P.-Y. Qian, K.-L. Wang, M.-Y. Wei, Potent antifouling marine dihydroquinolin-2 (1H)-one-containing alkaloids from the gorgonian coral-derived fungus *Scopulariopsis* sp, *Mar. Biotechnol.* 17 (2015) 408–415.
- [38] S.S. Al-Lihaibi, A. Abdel-Lateff, W.M. Alarif, H.S. Alorfi, Y. Nogata, T. Okino, Environmentally friendly antifouling metabolites from red sea organisms, *J. Chem.* 2019 (2019).
- [39] S. Qi, S. Zhang, L. Yang, P. Qian, Antifouling and antibacterial compounds from the gorgonians *Subergorgia suberosa* and *Scripseria gracilis*, *Nat. Prod. Res.* 22 (2008) 154–166.
- [40] A.S. Clare, Marine natural product antifoulants: status and potential, *Biofouling* 9 (1996) 211–229.
- [41] A.R. Davis, N.M. Targett, O.J. McConnell, C.M. Young, Epibiosis of marine algae and benthic invertebrates: natural products chemistry and other mechanisms inhibiting settlement and overgrowth, in: *Bioorg. Mar. Chemistry*, Springer, 1989, pp. 85–114.
- [42] L. Walters, M. Hadfield, C. Smith, Waterborne chemical compounds in tropical macroalgae: positive and negative cues for larval settlement, *Mar. Biol.* 126 (1996) 383–393.
- [43] D. Rittschof, R. Forward Jr., G. Cannon, J. Welch, M. McClary Jr., E. Holm, A. Clare, S. Conova, L. McKelvey, P. Bryan, Cues and context: larval responses to physical and chemical cues, *Biofouling* 12 (1998) 31–44.
- [44] D. Rittschof, Natural product antifoulants and coatings development, *Mar. Chem. Ecol.* 1 (2001) 543.
- [45] Z. Wang, P.-L. Li, X.-C. Luo, Q. Wang, L. van Ofwegen, X.-L. Tang, G.-Q. Li, Terpenoids from the south China sea soft coral *Sinularia multiflora*, *Nat. Prod. Res.* (2019) 1–8.
- [46] C.-Y. Wang, K.-L. Wang, P.-Y. Qian, Y. Xu, M. Chen, J.-J. Zheng, M. Liu, C.-L. Shao, C.-Y. Wang, Antifouling phenyl ethers and other compounds from the invertebrates and their symbiotic fungi collected from the South China Sea, *Amb. Express* 6 (2016) 102.
- [47] R. Baier, The role of surface energy in thrombogenesis, *Bull. N. Y. Acad. Med.* 48 (1972) 257.
- [48] L. Chambers, J. Wharton, R. Wood, F. Walsh, K. Stokes, Techniques for the measurement of natural product incorporation into an antifouling coating, *Prog. Org. Coating* 77 (2014) 473–484.
- [49] R.F. Brady Jr., Properties which influence marine fouling resistance in polymers containing silicon and fluorine, *Prog. Org. Coating* 35 (1999) 31–35.
- [50] M. Farid, M.A. El-Sockary, A.M. El-Saeed, A.I. Hashem, O.M.A. Elenien, M.S. Selim, S. Park, An eco-friendly non-isocyanate polyurethane treated by CO<sub>2</sub> as flame retardant nanocomposite coating/ZrO<sub>2</sub>@ SiO<sub>2</sub>, *Mater. Res. Express* 6 (2019), 065042.
- [51] M.S. Selim, H. Yang, S.A. El-Safty, N.A. Fatthallah, M.A. Shenashen, F.Q. Wang, Y. Huang, Superhydrophobic coating of silicone/β-MnO<sub>2</sub> nanorod composite for marine antifouling, *Colloids Surf., A* 570 (2019) 518–530.
- [52] M.S. Selim, H. Yang, F.Q. Wang, N.A. Fatthallah, Y. Huang, S. Kuga, Silicone/ZnO nanorod composite coating as a marine antifouling surface, *Appl. Surf. Sci.* 466 (2019) 40–50.
- [53] H. Jin, T. Zhang, W. Bing, S. Dong, L. Tian, Antifouling performance and mechanism of elastic graphene-silicone rubber composite membranes, *J. Mater. Chem. B* 7 (2019) 488–497.
- [54] J. Dustebek, C. Kandemir-Cavas, S.F. Nitodas, L. Cavas, Effects of carbon nanotubes on the mechanical strength of self-polishing antifouling paints, *Prog. Org. Coating* 98 (2016) 18–27.
- [55] Y. Ji, Y. Sun, Y. Lang, L. Wang, B. Liu, Z. Zhang, Effect of cnt/pdms nanocomposites on the dynamics of pioneer bacterial communities in the natural biofilms of seawater, *Materials* 11 (2018) 902.
- [56] E. Jin, Experimental Study on Antibacterial and Antifouling Strategy of Single Sarcophyton Trocheliophorum, Jilin University, 2017.
- [57] P. Bessell-Browne, R. Fisher, A. Duckworth, R. Jones, Mucous sheet production in Porites: an effective bioindicator of sediment related pressures, *Ecol. Indic.* 77 (2017) 276–285.
- [58] G. Nyland, H. Pavia, Chemical versus mechanical inhibition of fouling in the red alga *Dilsea carnosa*, *Mar. Ecol.: Prog. Ser.* 299 (2005) 111–121.
- [59] T. Masaki, D. Fujita, N. Hagen, The surface ultrastructure and epithallium shedding of crustose coralline algae in an 'Isokaye' area of southwestern Hokkaido, Japan, in: *Eleventh International Seaweed Symposium*, Springer, 1984, pp. 218–223.
- [60] K. Yamamoto, H. Endo, S. Yoshikawa, K. Ohki, M. Kamiya, Various defense ability of four sargassacean algae against the red algal epiphyte *Neosiphonia harveyi* in Wakasa Bay, Japan, *Aquat. Bot.* 105 (2013) 11–17.
- [61] R. Ganguli, V. Mehrotra, B. Dunn, Bioinspired living skins for fouling mitigation, *Smart Mater. Struct.* 18 (2009) 104027.
- [62] Y. Chen, Z. Liu, S. Han, J. Han, D. Jiang, Poly(propylene carbonate) polyurethane self-polishing coating for marine antifouling application, *J. Appl. Polym. Sci.* 133 (2016) 43667.
- [63] Y. Wang, X. Yao, S. Wu, Q. Li, J. Lv, J. Wang, L. Jiang, Bioinspired solid organogel materials with a regenerable sacrificial alkane surface layer, *Adv. Mater.* 29 (2017) 1700865.
- [64] C. Ma, L. Xu, W. Xu, G. Zhang, Degradable polyurethane for marine anti-biofouling, *J. Mater. Chem. B* 1 (2013) 3099–3106.
- [65] B. Ou, M. Chen, Y. Guo, Y. Kang, Y. Guo, S. Zhang, J. Yan, Q. Liu, D. Li, Preparation of novel marine antifouling polyurethane coating materials, *Polym. Bull.* 75 (2018) 5143–5162.
- [66] Y. Li, R. Chen, Y. Feng, L. Liu, X. Sun, L. Tang, K. Takahashi, J. Wang, Antifouling behavior of self-renewal acrylate boron polymers with pyridine-diphenylborane side chains, *New J. Chem.* 42 (2018) 19908–19916.
- [67] G. Dai, Q. Xie, C. Ma, G. Zhang, Biodegradable poly (ester-co-acrylate) with antifoulant pendant groups for marine anti-biofouling, *ACS Appl. Mater. Interfaces* 11 (2019) 11947–11953.
- [68] X. Zhou, Q. Xie, C. Ma, Z. Chen, G. Zhang, Inhibition of marine biofouling by use of degradable and hydrolyzable silyl acrylate copolymer, *Ind. Eng. Chem. Res.* 54 (2015) 9559–9565.
- [69] A. Salih, A. Larkum, G. Cox, M. Kühl, O. Hoegh-Guldberg, Fluorescent pigments in corals are photoprotective, *Nature* 408 (2000) 850–853.
- [70] M.P. Lesser, C.H. Mazel, M.Y. Gorbunov, P.G. Falkowski, Discovery of symbiotic nitrogen-fixing cyanobacteria in corals, *Science* 305 (2004) 997–1000.
- [71] S. Cao, J. Wang, D. Chen, Influence of illumination on settlement of diatom *Navicula* sp, *Microb. Ecol.* 62 (2011) 931–940.
- [72] M. Vesik, S. Jeffrey, Effect of blue-green light on photosynthetic pigments and chloroplast structure in unicellular marine algae from six classes, *J. Phycol.* 13 (1977) 280–288.
- [73] D.R. Bellwood, M.S. Pratchett, T.H. Morrison, G.G. Gurney, T.P. Hughes, J.G. Alvarez-Romero, J.C. Day, R. Grantham, A. Grech, A.S. Hoey, G.P. Jones, J.M. Pandolfi, S.B. Tebbett, E. Techera, R. Weeks, G.S. Cumming, Coral reef conservation in the Anthropocene: confronting spatial mismatches and prioritizing functions, *Biol. Conserv.* 236 (2019) 604–615.